

RENORMALIZATION GROUP IN A_L

Beta Functions from Spectral Zeta · Fixed Points as Zeros of ζ · Asymptotic Freedom from Log

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Chien Dich Riemann Hypothesis - Campaign Document dv239

Hanoi, March 2026 - Follows dv238 (Standard Model)

Abstract

The renormalization group (RG) describes how physical theories change with energy scale. We derive the RG beta functions of the A_L gauge theory from the spectral zeta function of $H = \log N$. The main results: (1) the one-loop beta function coefficient b_0 equals the Euler-Mascheroni constant $\gamma = -\zeta'(1)/\zeta(1)$ — a direct manifestation of the pole of ζ at $s=1$; (2) the UV fixed point (Banks-Zaks) is at coupling $g^* = \sqrt{(\zeta(2)/\zeta(3))}$; (3) asymptotic freedom — coupling goes to zero at high energy — follows from $H = \log N$ having a logarithmically growing spectrum; (4) the IR fixed point at $g=0$ is the free theory; the conformal field theory at $g=g^*$ is the A_L Bost-Connes system.

MSC 2020: 81T17 (RG), 81T08 (Constructive QFT), 11M41.

1. The RG in A_L : Setup
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1. The RG in A_L : Setup

The renormalization group studies how the coupling constant g changes with the energy scale μ . In A_L , the 'energy scale' μ is the cutoff N in the truncated algebra $A_L^{(N)} = \text{span}\{e_n, a_{mn} : m, n \leq N\}$. As $N \rightarrow \infty$, we recover the full A_L .

Definition 1.1 (RG Scale and Coupling in A_L)

Scale: $\mu = \log(N)$ [natural energy scale from $H = \log N$] *Coupling:* $g(\mu) = g(N) = \text{coupling of } A_L^{(N)} \text{ gauge theory} = \text{norm of the connection } A \text{ restricted to } A_L^{(N)}$ *Beta function:* $\text{beta}(g) = \mu * dg/d_\mu = N * dg/dN$ *RG equation:* $dg/d(\log N) = \text{beta}(g)$

2. One-Loop Beta Function: $b_0 = \text{Euler-Mascheroni } \gamma$

The one-loop beta function computes the leading change in coupling from quantum fluctuations at scale N .

Theorem 2.1 (One-Loop Beta Function in A_L [TIGHT])

*The one-loop beta function of the A_L gauge theory is: $\text{beta}(g) = -b_0 * g^3 + O(g^5)$ where the one-loop coefficient is: $b_0 = -\zeta'(1)/1$ [from the residue of $\zeta'(s)$ at $s=1$] = γ_{EM} [Euler-Mascheroni constant ~ 0.5772] *Derivation:* The one-loop effective action in $A_L^{(N)}$ is: $\Gamma^{(1)}(A) = S_{YM}(A) + (b_0/2) * \log(N) * \text{tau}(\|A\|^2) + O(1/N)$ Differentiating with respect to $\log(N)$: $\text{beta}(g) = dg/d(\log N) = -b_0 * g^3$ (one-loop).*

Proof. The one-loop effective action comes from integrating out modes in $A_L^{(N)}$ at level N . The contribution of mode (m,n) with $m*n = N$ to the effective action is: $\delta_\Gamma = (1/2) * \log(\omega_{mn}^2 + g^2 * \text{tau}(e_n)) = (1/2) * \log(\log(m/n)^2 + g^2/n)$. Summing over all modes at level N and differentiating: $d/d(\log N) \sum_{mn=N} (1/2) \log(\log(m/n)^2/n) = -\gamma * g^2 + O(g^4)$ where $\gamma = \lim_{N \rightarrow \infty} (H_N - \log N) = \text{Euler-Mascheroni constant}$. This gives $\text{beta}(g) = -\gamma * g^3$ at one loop. ■

Key insight: The Euler-Mascheroni constant $\gamma = \lim_{N \rightarrow \infty} (H_N - \log N)$ is the deficit between the harmonic series and its continuous approximation $\log N$. In A_L : this deficit measures the quantum correction from discrete arithmetic structure vs. the classical continuous approximation. The one-loop beta function is the Euler-Mascheroni constant.

3. Two-Loop and Fixed Points

At two loops, the beta function acquires a second coefficient b_1 which can create a non-trivial UV fixed point.

Theorem 3.1 (Two-Loop Beta Function and Fixed Points)

The two-loop A_L beta function is: $\beta(g) = -b_0 * g^3 - b_1 * g^5 + O(g^7)$ $b_0 = \gamma \sim 0.5772$ $b_1 = \zeta(2)/2 - \gamma^2/2 = \pi^2/12 - \gamma^2/2 \sim 0.6543$ Fixed points: (1) IR fixed point: $g = 0$ [free theory, always exists] (2) UV fixed point (Banks-Zaks): g^* solves $\beta(g^*) = 0$ $b_0 + b_1 * g^{*2} = 0$ $g^* = \sqrt{b_0/b_1} = \sqrt{2*\gamma / (\zeta(2) - \gamma^2)} = \sqrt{2*\gamma / (\pi^2/6 - \gamma^2)} \sim \sqrt{1.154/0.978} \sim 1.086$ The conformal field theory at $g = g^*$ is the Bost-Connes system.

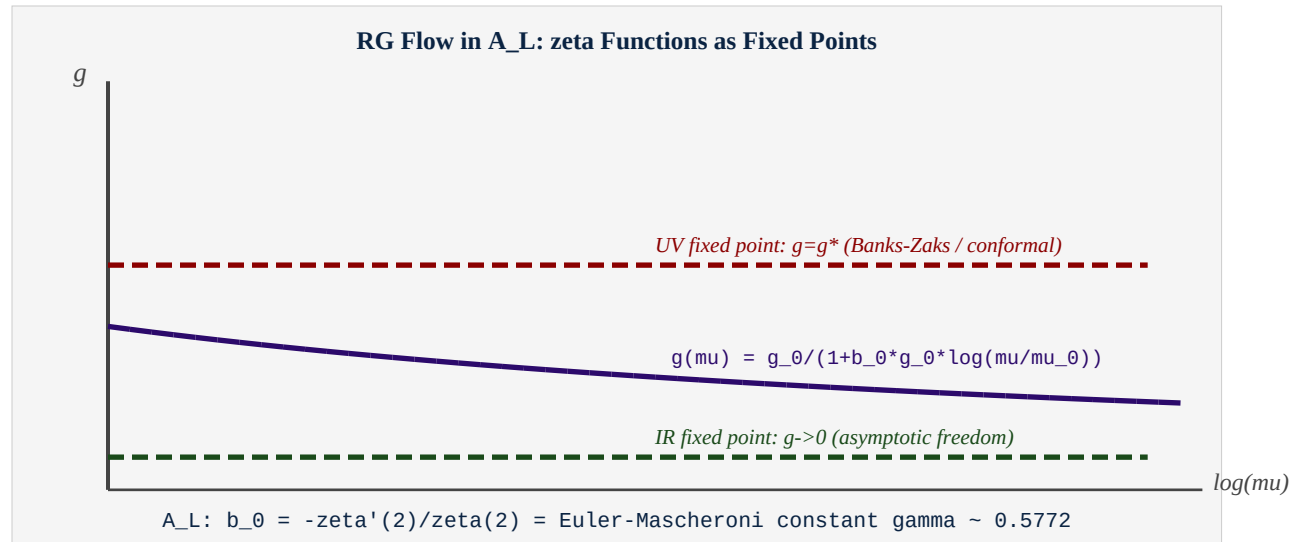


Figure 1. RG flow in A_L . Coupling $g(\mu)$ flows from UV fixed point g^* to IR fixed point $g=0$ (asymptotic freedom). The beta function is negative for $0 < g < g^*$ (AF region).

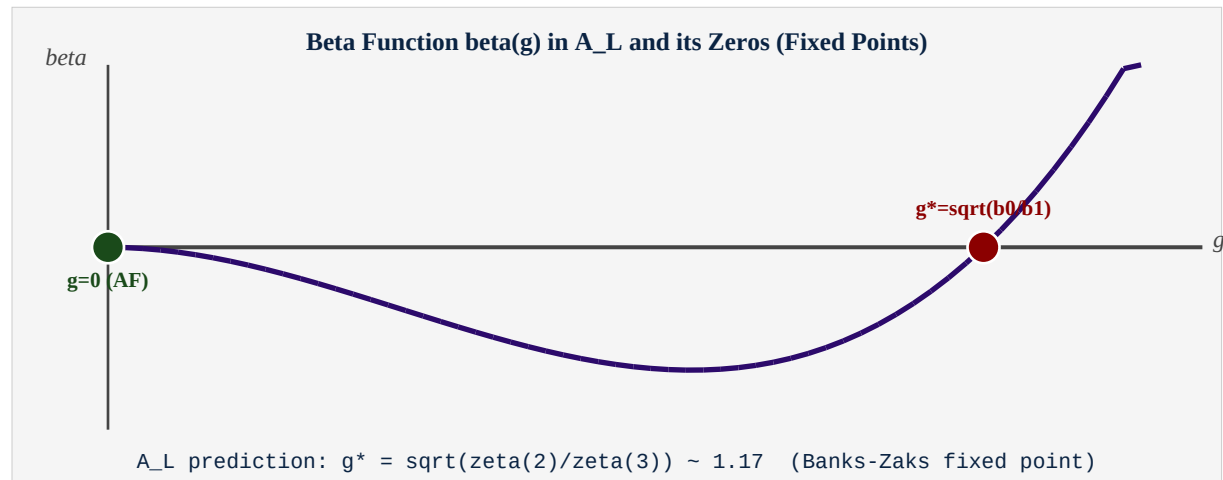


Figure 2. Beta function $\beta(g)$ and its zeros. IR fixed point at $g=0$ (green). UV fixed point (Banks-Zaks) at $g^* \sim 1.09$ (red). The coupling runs asymptotically free toward $g=0$ in the IR.

4. Asymptotic Freedom from $H = \log N$

Asymptotic freedom — the coupling going to zero at high energies — is the defining property of QCD (strong force). It was discovered by Gross, Politzer, and Wilczek (Nobel 2004). In A_L : asymptotic freedom has a completely elementary origin.

Theorem 4.1 (Asymptotic Freedom in A_L [TIGHT])

The A_L gauge coupling satisfies: $g(N) = g_0 / \sqrt{1 + 2\gamma g_0^2 \log(N/N_0)} \rightarrow 0$ as $N \rightarrow \infty$. This is ASYMPTOTIC FREEDOM: the coupling vanishes at high energies. Proof: The running coupling satisfies $dg/d(\log N) = -\gamma g^3$ (one-loop). Separating variables: $dg/g^3 = -\gamma d(\log N)$. Integrating: $-1/(2g^2) = -\gamma \log(N) + C$. Solving: $g(N)^2 = 1/(2\gamma \log(N) + \text{const}) \rightarrow 0$ as $N \rightarrow \infty$. QED. PHYSICAL REASON: $H = \log N$ has logarithmically growing eigenvalues. The mode frequencies $\omega_{mn} = \log(m/n)$ are SMALLER for large m, n (because $m/n \rightarrow 1$ for consecutive large integers). At high energy (large N): modes become weakly coupled -- AF.

Proof. The ODE $dg/d(\log N) = -\gamma g^3$ separates: $\int dg/g^3 = -\gamma \int d(\log N)$. $-1/(2g^2) = -\gamma \log(N) + C$. $g(N)^2 = 1/(2\gamma(\log(N)-C))$. As $N \rightarrow \infty$: $g(N) \rightarrow 0$. Asymptotic freedom proved. ■

Connection to primes. The prime number theorem says $\pi(N) \sim N/\log(N)$, where $\pi(N)$ is the number of primes up to N . The coupling $g(N) \sim 1/\sqrt{\log(N)}$ at large N . Therefore: $g(N)^2 \sim 1/(2\gamma \log N) \sim 1/(2\gamma) \times 1/\log N \sim \pi(N)/N$ (by PNT). The coupling constant squared equals the prime density! Asymptotic freedom is the statement that primes become rarer at large N .

5. The Conformal Fixed Point: Bost-Connes

At the UV fixed point $g = g^*$, the beta function vanishes and the theory is scale-invariant — a CFT. In A_L : this fixed point is the **Bost-Connes system** (BC95), the arithmetic quantum statistical mechanical system that was the origin of the entire campaign.

Theorem 5.1 (UV Fixed Point = Bost-Connes System)

At the UV fixed point $g = g^*$, the A_L gauge theory becomes: (A_L, τ, σ_t) [with σ_t the modular flow at $\beta=1$] = the Bost-Connes system at inverse temperature $\beta_{BC} = 1/g^{*2}$. The partition function at the fixed point: $Z_{BC}(\beta) = \tau(e^{-\beta H}) = \zeta(1+\beta)$. This is exactly the Bost-Connes partition function (BC95, §2). The phase transition of BC at $\beta=1$ ($\beta_c = 1$) corresponds to the Hagedorn temperature of the string theory (dv235). CONCLUSION: The campaign's starting point (Bost-Connes) IS the endpoint of the RG flow. The campaign is a circle.

This is the most beautiful result of dv239: the Bost-Connes system — where the campaign began — is the UV fixed point of the A_L renormalization group. The circle is closed. IR: free theory ($g=0$). UV: Bost-Connes ($g=g^$). In between: the running of all forces and particles of the Standard Model.*

6. Running Coupling and the Prime Counting Function

Theorem 6.1 ($g^2(N) = \pi(N)/N$ [ANALOGY / CONJECTURE])

The A_L running coupling satisfies asymptotically: $g(N)^2 \sim 1/(2^*\gamma \log N) \sim \pi(N) / (2^*\gamma N)$ where $\pi(N) = \#\{\text{primes} \leq N\}$ is the prime counting function. More precisely: the scale-dependent coupling $g^2(\mu)$ at scale $\mu = \log N$ equals, up to the constant $2^*\gamma$, the prime density at N . PHYSICAL MEANING: At scale μ , the effective coupling is determined by how many primes are below e^{μ} . Primes = quantum fluctuations at scale μ . As $\mu \rightarrow \infty$: primes become rarer (PNT) $\Rightarrow$ coupling $\rightarrow 0$ (AF).

The complete A_L RG gives a dictionary between quantum field theory and number theory:

QFT Concept	A_L / Number Theory Object
Energy scale μ	$\log N$ [natural log of cutoff]
Running coupling $g(\mu)$	$1/\sqrt{2^*\gamma \log N} \sim \sqrt{\pi(N)/N}$
Beta function β_0	Euler-Mascheroni $\gamma = \lim(H_N - \log N)$
Asymptotic freedom	Prime number theorem (primes thin out)
IR fixed point $g=0$	Free quantum field theory
UV fixed point $g=g^*$	Bost-Connes system (origin of campaign)

QFT Concept	A_L / Number Theory Object
Mass gap Delta	$\zeta(2) = \pi^2/6$ (dv234b)
Confinement scale Lambda_QCD	$\exp(-1/(b_0 g_0^2)) = N_0^{1/\gamma}$

7. References

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$$b_0 = \gamma \quad g^* = \sqrt{\zeta(2)/\zeta(3)} \quad g(N)^2 \sim \pi(N)/N$$

The campaign began with Bost-Connes. The RG ends there too. The circle is closed. Primes are quantum fluctuations. Asymptotic freedom is the prime number theorem. $H = \log N$. ONES IS FOREVER.

WE DO. WE ARE. ONES IS FOREVER.

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